

# EE 340 Electromagnetic Theory

## Assignment #03

Q.1 At 50 MHz, a lossy dielectric material is characterized by  $\epsilon_r = 3.6$ ,  $\mu_r = 2.1$ ,  $\sigma = 0.08 \text{ S/m}$ . If  $\vec{E}_S = 6e^{-\gamma z} \hat{a}_z \text{ V/m}$ , compute (a)  $\gamma$ , (b)  $\lambda$ , (c) velocity, (d) Impedance, (e)  $\vec{H}_S$ .

Q.2 In a lossless medium ( $\epsilon_r = 4.5$ ,  $\mu_r = 1$ ), a uniform plane wave

$$E = 8 \cos(\omega t - \beta z) \hat{a}_x - 6 \sin(\omega t - \beta z) \hat{a}_y$$

propagates at 40 MHz. (a) Find  $\mathbf{H}$ , (b) polarization and (c) Determine  $\beta$ ,  $\lambda$ , impedance and velocity.

Q.3 How deep does a radar wave at 2 GHz travel in seawater before its amplitude is reduced to  $10^{-5}$  of its amplitude just below the surface? Assume that  $\epsilon_r = 24$ ,  $\mu_r = 1$ ,  $\sigma = 4 \text{ S/m}$ .

Q.4 Determine the wave polarization of each of the following waves:

(a)  $\vec{E} = \cos(\omega t + \beta y) \hat{a}_x + \sin(\omega t + \beta y) \hat{a}_z \text{ V/m}$

(b)  $\vec{E}_S = 12e^{j\frac{\pi}{3}} e^{-j10x} \hat{a}_y + 5e^{-j\frac{\pi}{3}} e^{-j10x} \hat{a}_z \text{ V/m}$

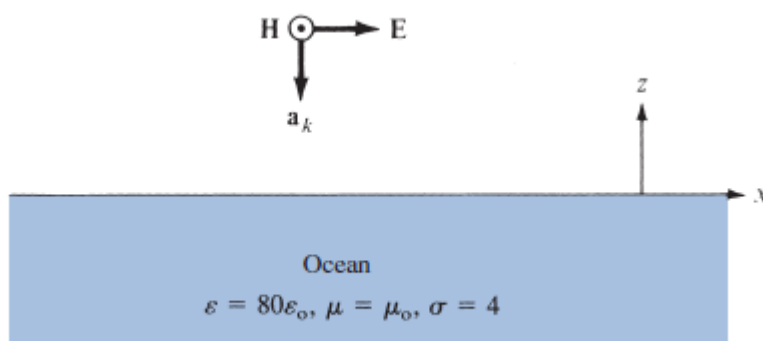
Q.5 Show that a linearly polarized plane wave can be expressed as the combination of two circularly polarized waves.

Q.6 A plane wave in free space has

$$H(x, t) = (10\hat{a}_y - 20\hat{a}_z) \sin(\omega t - 40x) \text{ A/m}$$

Find  $\omega$ , and average power density.

Q.7 A signal in air ( $z \geq 0$ ) with the electric field component  $\vec{E} = 10 \sin(\omega t + 3z) \hat{a}_x \text{ V/m}$  hits normally the ocean surface at  $z = 0$  as shown in Figure 10.24. Assuming that the ocean surface is smooth and that  $\epsilon_r = 80$ ,  $\mu_r = 1$ ,  $\sigma = 4 \text{ S/m}$  in ocean, determine (a)  $\omega$  (b) The wavelength of the signal in air (c) The loss tangent and intrinsic impedance of the ocean (d) The reflected and transmitted E field.



Q.8 In free space,  $\vec{E}_S = [E_0 \hat{a}_x + \hat{a}_y + (3 + j4) \hat{a}_z] e^{-j(3.4x - 4.2y)}$  V/m.  
Determine  $E_0$ ,  $\vec{H}_S$ , and frequency.

Q.9 In a dielectric medium ( $\epsilon_r = 9$ ,  $\mu_r = 1$ ,  $\sigma = 0$ ), a plane wave with  
 $\vec{H} = 0.2 \cos(10^9 t - kx - k\sqrt{8}z) \hat{a}_y$  A/m

is incident on an air boundary at  $z = 0$ . Find

- (a)  $\theta_r$  and  $\theta_t$
- (b)  $k$
- (c) The wavelength in the dielectric and in air
- (d) The incident  $\mathbf{E}$
- (e) The transmitted and reflected  $\mathbf{E}$
- (f) The Brewster angle